

**PRAKTIKUM SISTEM OPERASI**

**MODUL IV**

**MEMBACA SOURCE CODE XINU**



Oleh:

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**PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK**

**DIREKTORAT KAMPUS PURWOKERTO**

**UNIVERSITAS TELKOM**

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# JURNAL

## 1. Soal 1

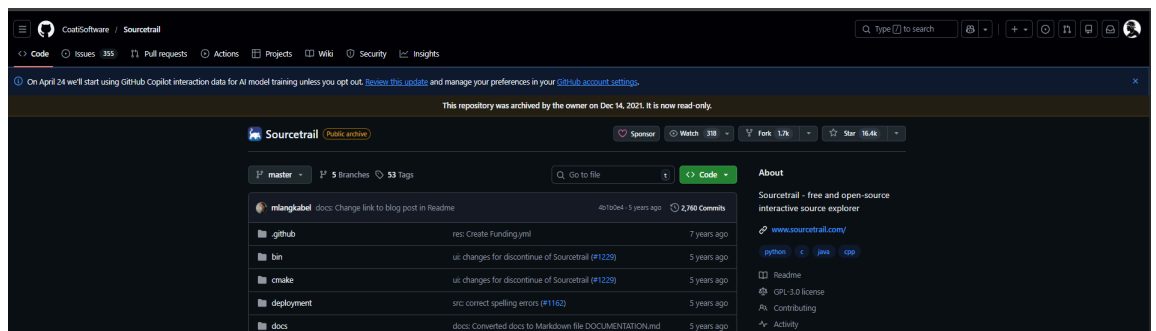
Apa nama image yang dihasilkan setelah melakukan kompilasi pada Xinu? Berapa ukuran file tersebut? Ada pada folder apa file image tersebut?

- File image yang dihasilkan adalah xinu.elf, file tersebut berada di dalam folder xinu/compile/, yaitu direktori yang sama dengan tempat proses kompilasi dilakukan, Untuk mengetahui ukuran file tersebut, dapat digunakan perintah `ls -lh xinu.elf` pada terminal Linux setelah proses make selesai. Ukuran file image ini umumnya berkisar antara 300 KB hingga 1 MB

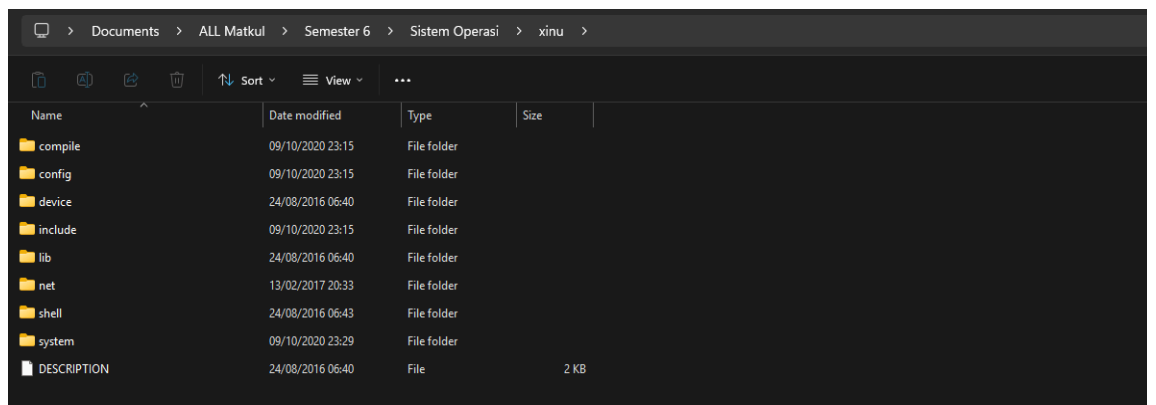
## 2. Soal 2

Membaca source code Xinu

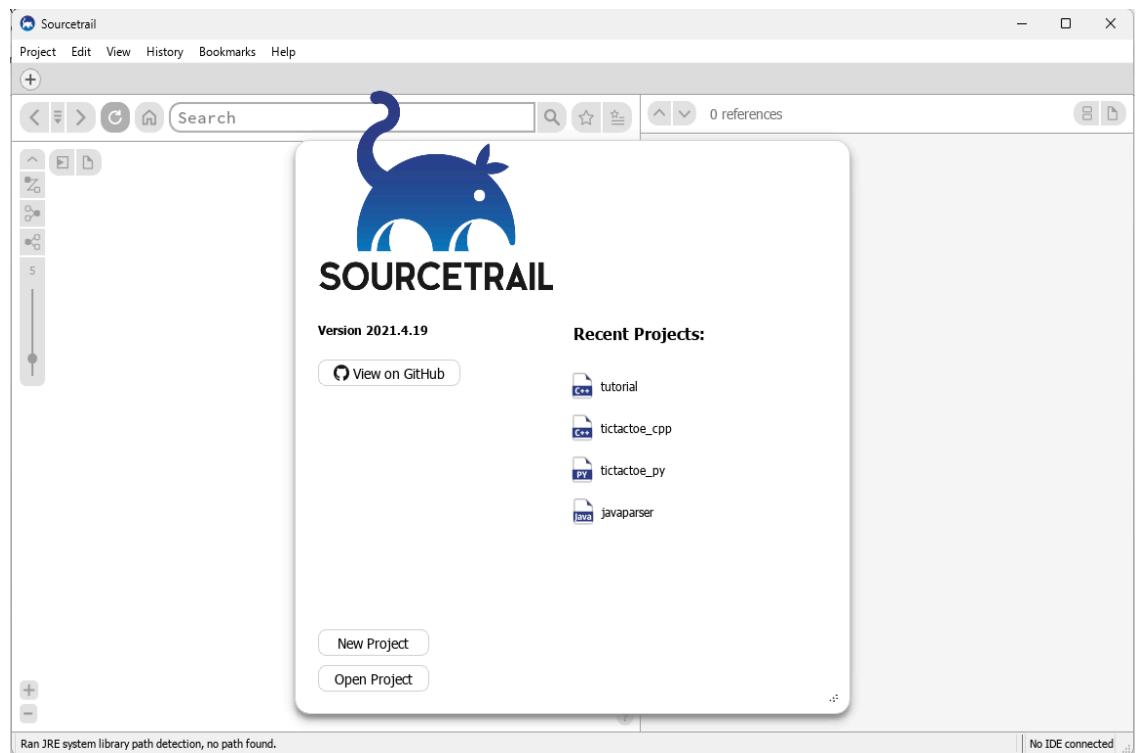
- Cek aplikasi bernama Sourcetrail di PC. SourceTrail adalah software untuk mengeksplorasi source code. Programmer yang bagus lebih banyak membaca kode daripada menulis kode



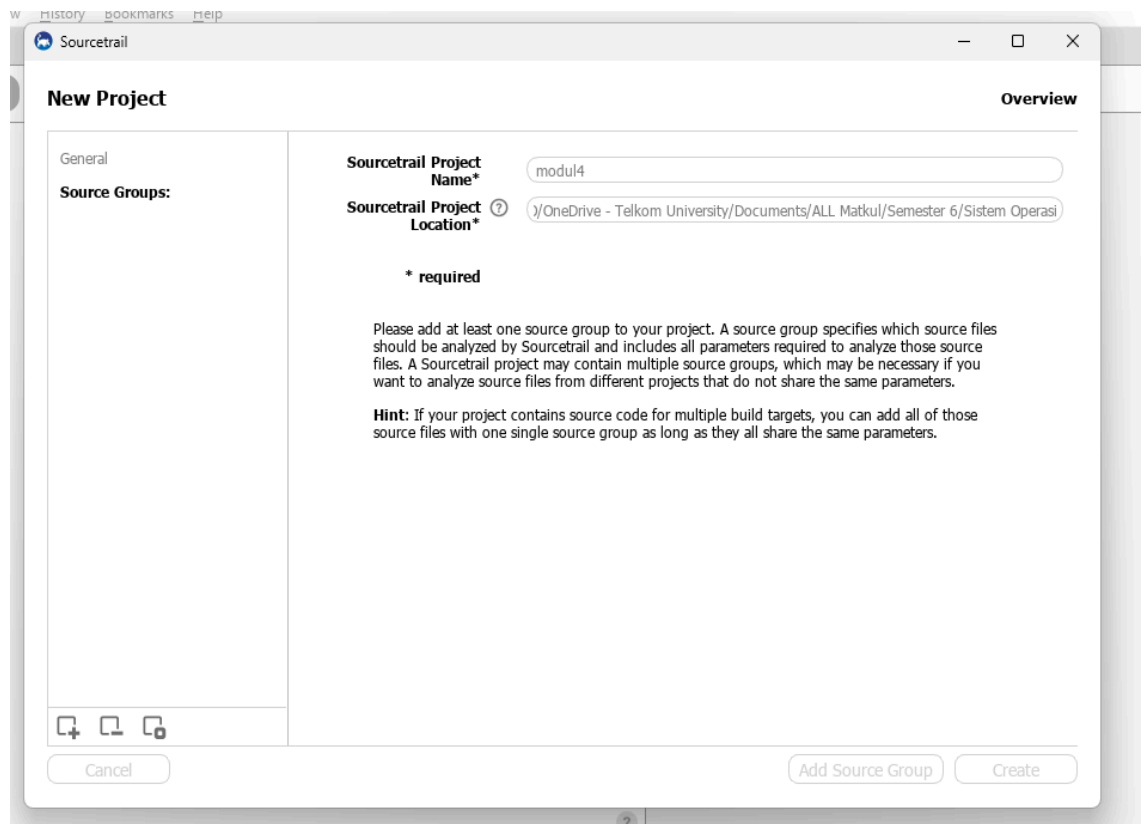
- Download file source code xinu yang tersedia pada attempt jurnal praktikum di LMS



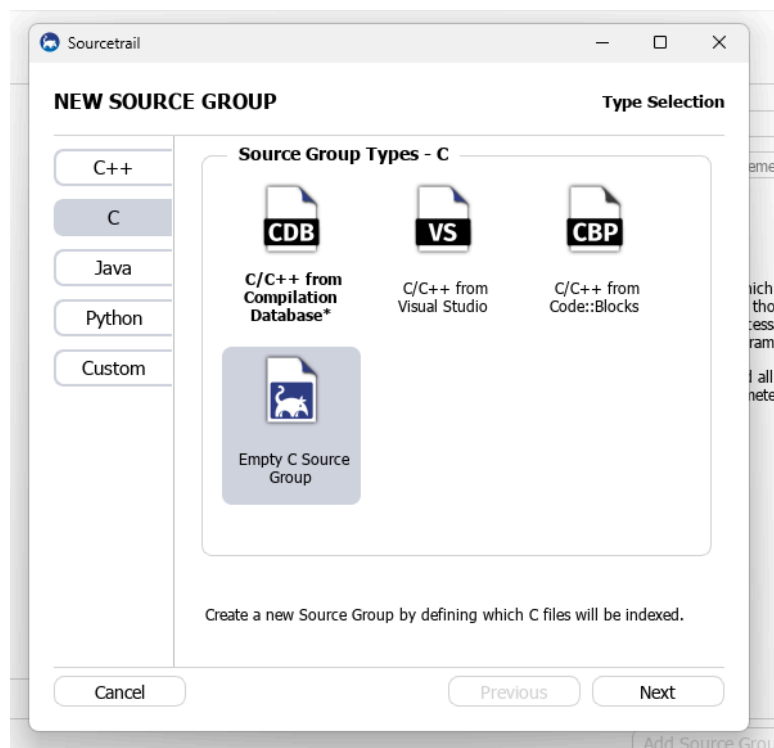
- Jalankan SourceTrail



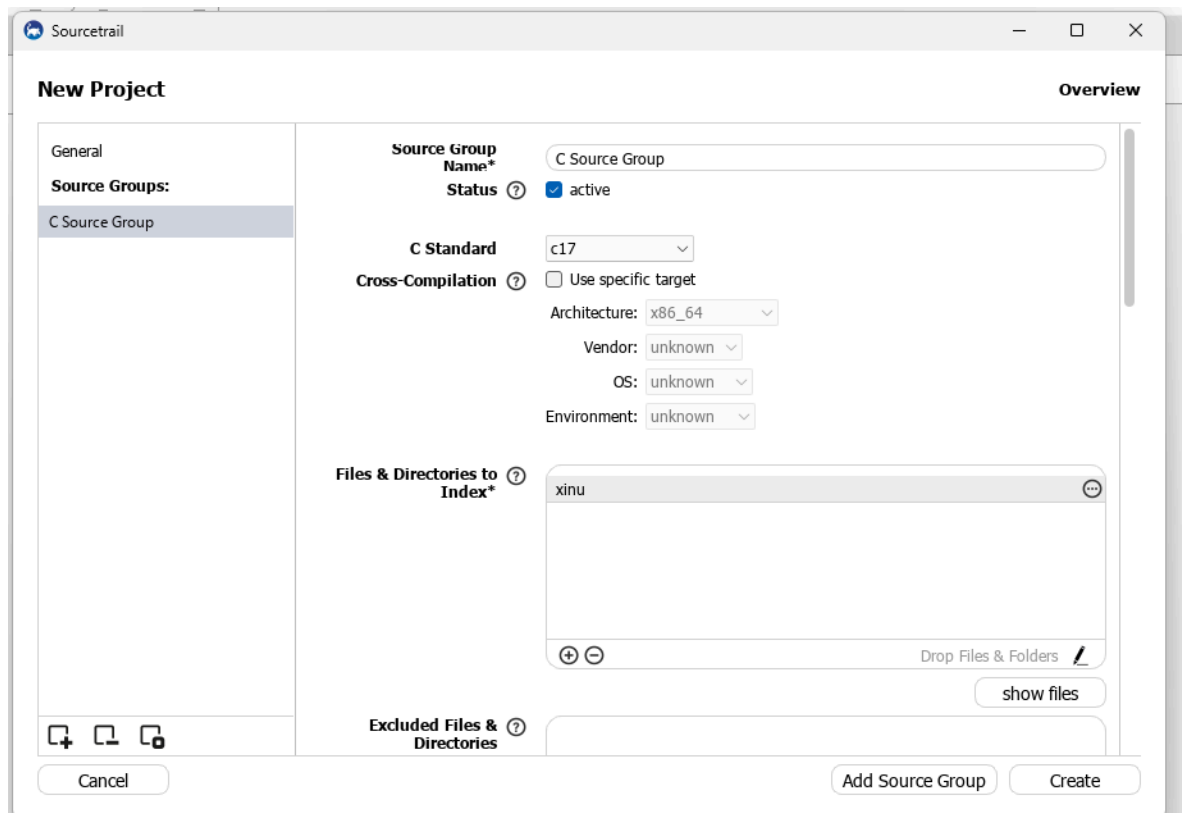
- d. Project → New Project, Isi nama project xinu dan pilih lokasi project di manapun



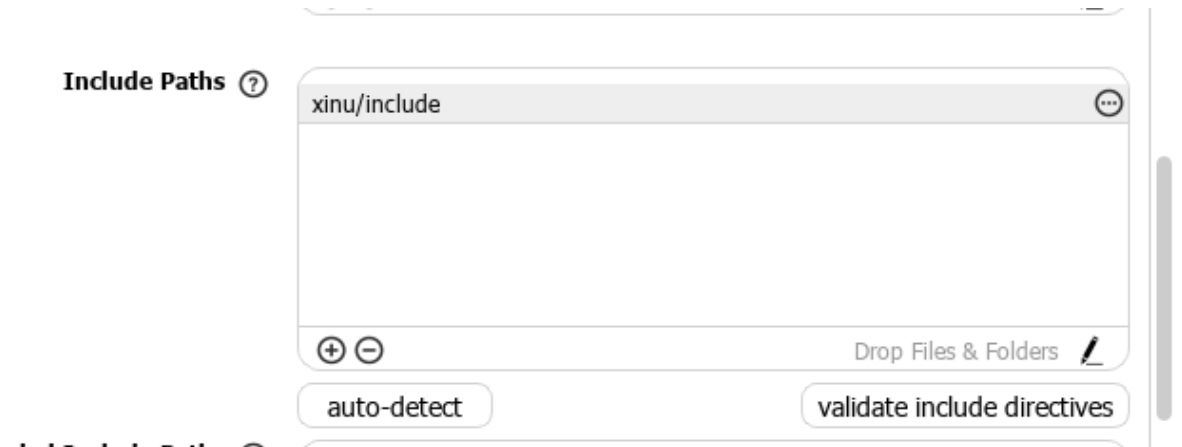
- e. Add Source Groups, pilih C, lalu pilih Empty C Source Group



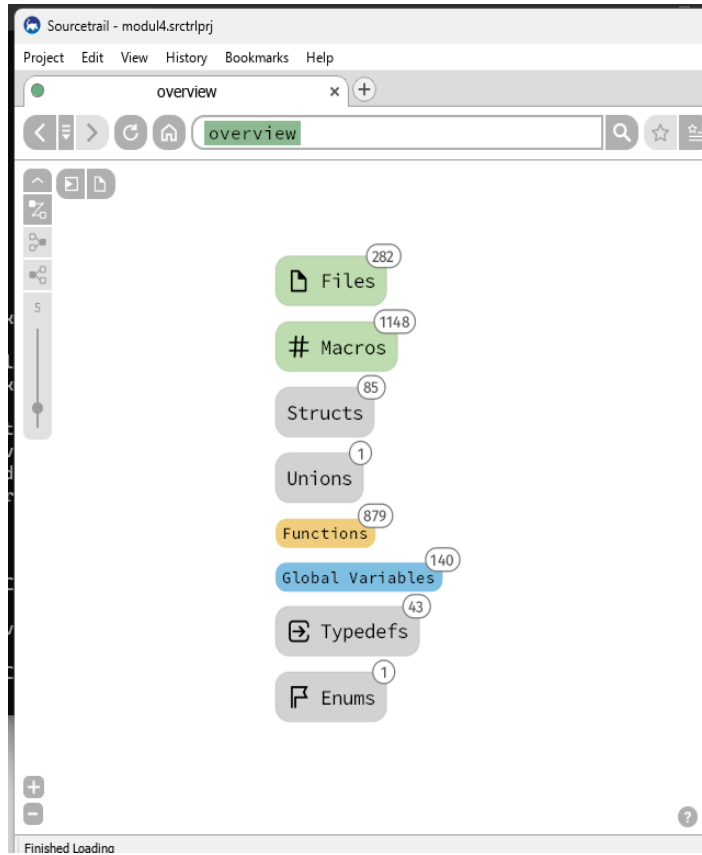
- f. File & Directories to Index: masukkan semua folder Xinu (yang sebelumnya telah di download)



- g. Include Paths: .../xinu/include

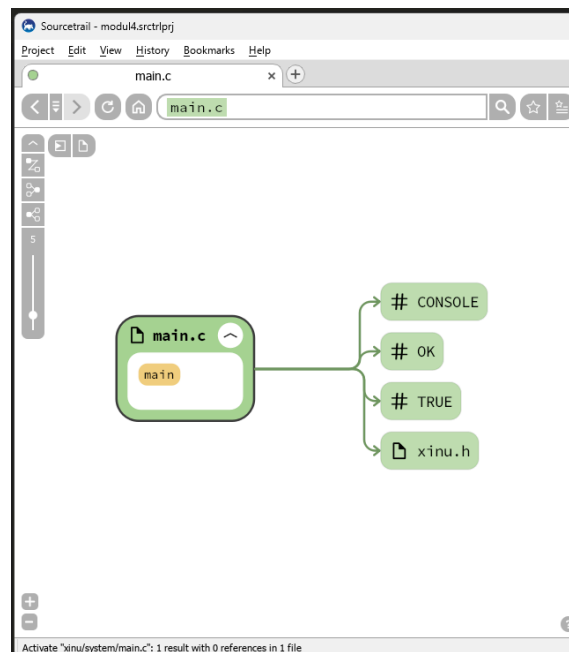


h. Create

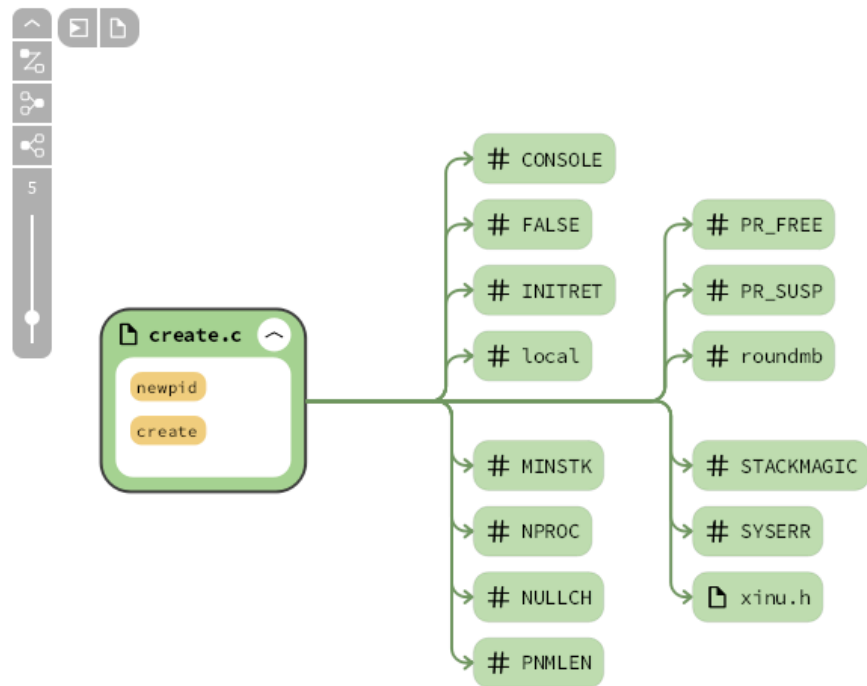


i. Silahkan eksplorasi source code Xinu

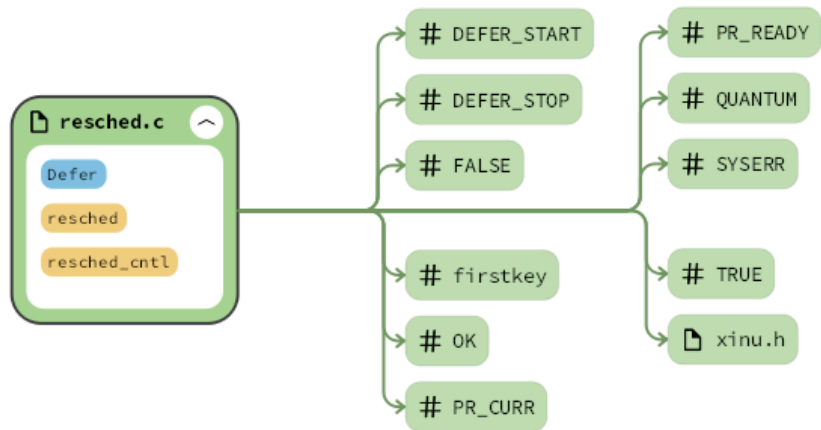
- main



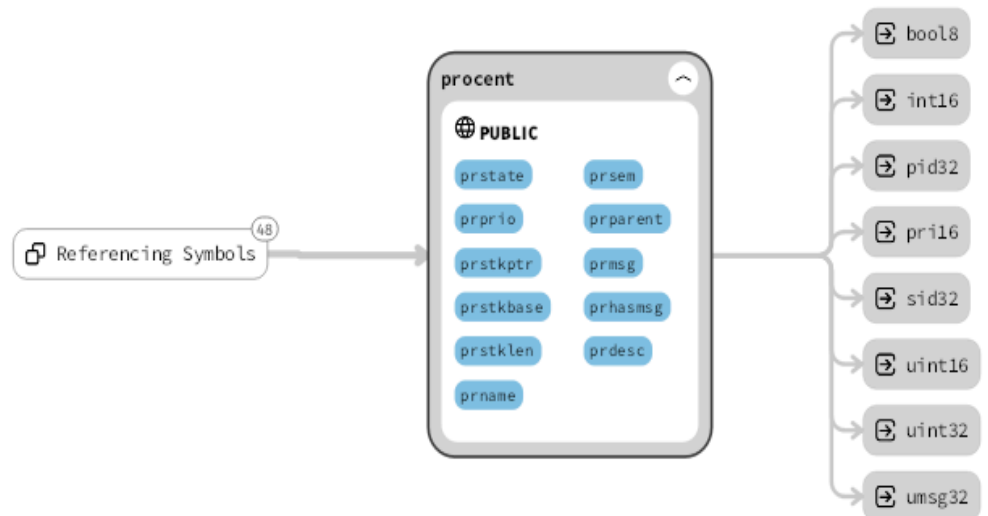
- create



- resched



- procent



### 3. Soal 3

Carilah struktur data dari proses pada Xinu OS. Struktur data proses ada pada file apa? Informasi apa saja yang disimpan dalam struktur data tersebut?

- Struktur data untuk menemukan process.h terletak pada directory berikut:

```
xinu@xinu-develop-end:~$ find / -name "process.h" 2>/dev/null
/home/xinu/prev-version/include/process.h
/home/xinu/xinu/include/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/internal/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/iprt/process.h
/usr/src/virtualbox-guest-4.3.28/include/internal/process.h
/usr/src/virtualbox-guest-4.3.28/include/iprt/process.h
```

- Setelah menemukan dimana directory file process.h saya menggunakan perintah “grep -r "struct procent" /home/xinu/xinu/include/” ditemukan strukturnya di file “process.h” dan sudah terbukti hasilnya seperti berikut:



```

xinu@xinu-develop-end:~$ grep -r "struct procent" /home/xinu/xinu/include/
/home/xinu/xinu/include/process.h:struct procent {                /* Entry in the process
table                */
xinu@xinu-develop-end:~$ cat /home/xinu/xinu/include/process.h
/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC      8
#endif

/* Process state constants */

#define PR_FREE      0      /* Process table entry is unused      */
#define PR_CURR      1      /* Process is currently running      */
#define PR_READY     2      /* Process is on ready queue          */
#define PR_RECV      3      /* Process waiting for message        */
#define PR_SLEEP     4      /* Process is sleeping                */
#define PR_SUSP      5      /* Process is suspended               */
#define PR_WAIT      6      /* Process is on semaphore queue      */
#define PR_RECTIM    7      /* Process is receiving with timeout  */

/* Miscellaneous process definitions */

#define PNMLEN      16      /* Length of process "name"          */
#define NULLPROC    0      /* ID of the null process              */

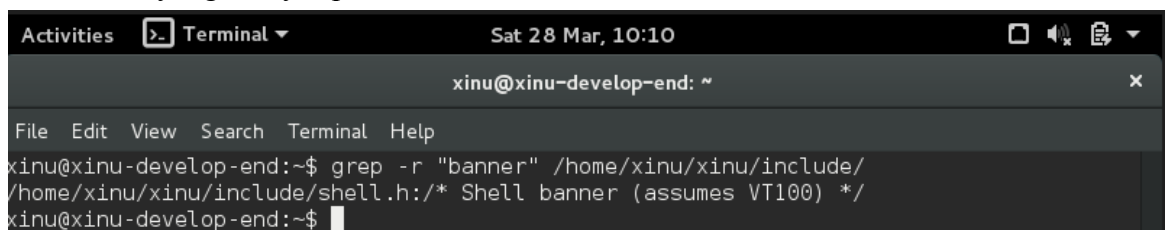
```

Hal ini dibuktikan dengan output terminal yang menampilkan struct procent { /\* Entry in the process table \*/.

#### 4. Soal 4

Mengubah welcome banner pada Xinu!

- Carilah file yang menyimpan banner Xinu!



```

Activities  Terminal  Sat 28 Mar, 10:10
xinu@xinu-develop-end: ~
File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ grep -r "banner" /home/xinu/xinu/include/
/home/xinu/xinu/include/shell.h:/* Shell banner (assumes VT100) */
xinu@xinu-develop-end:~$

```



```

xinu@xinu-develop-end:~$ cat /home/xinu/xinu/shell/shell.c
/* shell.c - shell */

#include <xinu.h>
#include <stdio.h>
#include "shprototypes.h"

/*****
/* Table of Xinu shell commands and the function associated with each */
*****/
const struct cmdent cmdtab[] = {
    {"argecho",    xsh_argecho},
    {"arp",        xsh_arp},
    {"cat",        xsh_cat},
    {"clear",      xsh_clear},
    {"date",       xsh_date},
    {"devdump",    xsh_devdump},
    {"echo",       xsh_echo},
    {"help",       xsh_help},
    {"ls",         xsh_ls},
    {"kill",       xsh_kill},
    {"memdump",    xsh_memdump},
    {"memstat",    xsh_memstat},
    {"ns",         xsh_ns},
    {"netinfo",    xsh_netinfo},
    {"ping",       xsh_ping},
    {"ps",         xsh_ps},
    {"sleep",      xsh_sleep},

```

Activities  Terminal ▾ Sat 28 Mar, 10:15 

xinu@xinu-develop-end: ~ 

File Edit View Search Terminal Help

```

        {"tee",        xsh_tee},
        {"udp",        xsh_udpdump},
        {"udpecho",    xsh_udpecho},
        {"udpeserver", xsh_udpeserver},
        {"uptime",     xsh_uptime},
        {"?",          xsh_help}
};

uint32 ncmd = sizeof(cmdtab) / sizeof(struct cmdent);

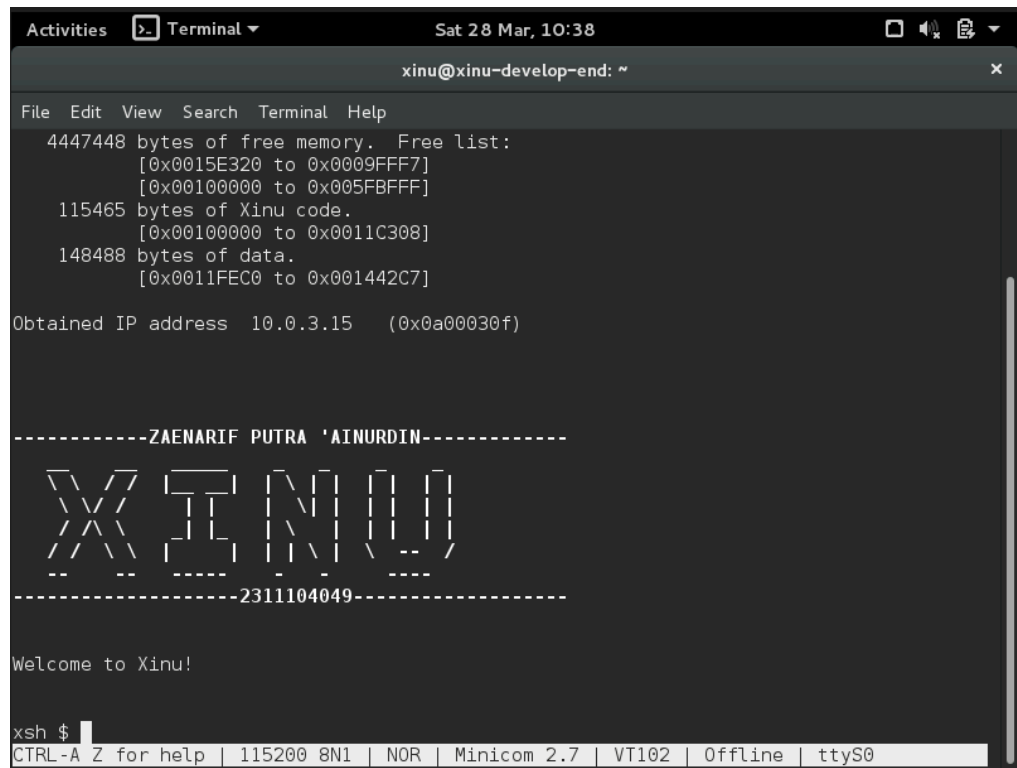
```

- 
- The screenshot shows a terminal window with the title bar "xinu@xinu-develop-end: ~". The terminal is running the GNU nano 2.2.6 text editor, editing the file "/home/xinu/xinu/include/shell.h". The editor's status bar at the top indicates "File: /home/xinu/xinu/include/shell.h" and "Modified".
- The content of the file being edited is a shell banner script. It starts with a comment "/\* Shell banner (assumes VT100) \*/". The script defines several shell variables in red text:
- `SHELL_BAN0`: Set to `"\033[31;1m"` (red, bold).
  - `SHELL_BAN1`: Set to `"-----ZAENARIF PUTRA 'AINURDIN-----"`.
  - `SHELL_BAN2`: Set to `"`.
  - `SHELL_BAN3`: Set to `"`.
  - `SHELL_BAN4`: Set to `"`.
  - `SHELL_BAN5`: Set to `"`.
  - `SHELL_BAN6`: Set to `"`.
  - `SHELL_BAN7`: Set to `"`.
  - `SHELL_BAN8`: Set to `"-----2311104049-----"`.
  - `SHELL_BAN9`: Set to `"\033[0m\n"` (reset color).
- The banner text is displayed in yellow on a black background. It features a stylized "XINU" logo in the center, composed of vertical and diagonal lines. The text "ZAENARIF PUTRA 'AINURDIN" is on the left and "2311104049" is on the right, separated by a large "XINU" logo.
- Below the banner definitions, there is a comment "/\* Messages shell displays for user \*/" followed by several more definitions in red text:
- `SHELL_PROMPT`: Set to `"xsh $ "` with a comment "/\* Command prompt \*/".
  - `SHELL_STRMSG`: Set to `"Welcome to Xinu!\n"` with a comment "/\* Welcome message \*/".
  - `SHELL_EXITMSG`: Set to `"Shell closed\n"` with a comment "/\* Shell exit message \*/".
  - `SHELL_SYNERMSG`: Set to `"Syntax error\n"` with a comment "/\* Syntax error message \*/".
  - `SHELL_CREATMSG`: Set to `"Cannot create process\n"` with a comment "/\* command error \*/".
  - `SHELL_INERRMSG`: Set to `"Cannot open file %s for input\n"` with a comment "/\* Input err \*/".
  - `SHELL_OUTERRMSG`: Set to `"Cannot open file %s for output\n"` with a comment "/\* Output err \*/".
- At the bottom of the terminal, there is a row of keyboard shortcuts for nano:
- `^G` Get Help
  - `^O` WriteOut
  - `^R` Read File
  - `^Y` Prev Page
  - `^K` Cut Text
  - `^C` Cur Pos
  - `^X` Exit
  - `^J` Justify
  - `^W` Where Is
  - `^V` Next Page
  - `^U` UnCut Text
  - `^T` To Spell

Untuk mengubah welcome banner pada Xinu, pertama dicari terlebih dahulu file yang menyimpan banner menggunakan perintah `grep -r "banner" /home/xinu/xinu/` sehingga ditemukan bahwa banner tersimpan di file `shell.h` dalam bentuk `define SHELL_BAN0` hingga `SHELL_BAN9` dan ditampilkan oleh file `shell.c`. Selanjutnya file `shell.h` dibuka menggunakan `nano` kemudian isi `define` banner diubah sesuai nama dan nim, dengan mengubah nama saya & nim saya yaitu : Zaenarif Putra 'A - 2311104049. Setelah perubahan disimpan, dilakukan kompilasi ulang menggunakan perintah `make clean` dan `make` di direktori `xinu/compile`, lalu dijalankan `sudo minicom` beserta VM backend sehingga banner baru yang memuat nama dan NIM berhasil tampil di layar.

d. Setelah melakukan perubahan lakukan perintah berikut ini pada terminal Linux

- \$ cd xinu/compile
- \$ make clean
- \$ make
- \$ sudo minicom
- Jalankan vm backend



The screenshot shows a terminal window titled "Terminal" with the date and time "Sat 28 Mar, 10:38". The user is logged in as "xinu@xinu-develop-end: ~". The terminal displays the following output:

```
File Edit View Search Terminal Help
4447448 bytes of free memory. Free list:
[0x0015E320 to 0x0009FFF7]
[0x00100000 to 0x005FBFFF]
115465 bytes of Xinu code.
[0x00100000 to 0x0011C308]
148488 bytes of data.
[0x0011FEC0 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)

-----ZAENARIF PUTRA 'AINURDIN-----

  X I N U
  _ _ _ _ _
-----2311104049-----

Welcome to Xinu!

xsh $
```

At the bottom of the terminal, there is a status bar with the text: "CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0".